Data Cleaning

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Our research question is "Is there a correlation between Air Quality Health Index and income inequality in Canadian metropolitan areas?"

We've collected our Air Quality Health Index (AQHI) data. We now need to identify the Census Metro Areas (CMAs) associated with each Station ID.

First, let's change our AQHI data to refer to a city rather than a station code!

```
library(tidyverse) #importing libraries, not all used yet
library(gtools)
library(modelsummary)
library(ggeffects)
library(data.table)
library(sf)
library(jsonlite)
library(data.table)
library(tmap)
library(ggplot2)
library(terra)
library(dplyr)
library(tidyr)
```

```r
main_dir <- "C:/Users/zaneb/Documents/Vault/UBC/24W/Term 2/ECON 326/FinalProject" #equivalent

regions <- c("atl", "ont", "pnr", "pyr") #the four areas of data collection in AQHI files

d_AQHI <- lapply(setNames(regions, regions), function(region) { #lapply applies a function to
  dir_path <- file.path(main_dir, "Data/AQHI", region) #folder of data
  files <- list.files(path = dir_path, pattern = "\\.csv$", full.names = TRUE) #list of files
  rbindlist(lapply(files, fread), use.names = TRUE, fill = TRUE) #reading all of the folders
})

naps <- st_read(file.path(main_dir, "Data/aqhi_station.geojson")) #read the geojson guide to
```

```
Reading layer `aqhi_station' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\aqhi_station.geoj
  using driver `GeoJSON'
Simple feature collection with 135 features and 5 fields
Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: -135.0808 ymin: 42.28306 xmax: -52.72528 ymax: 68.3609
Geodetic CRS:  WGS 84
```

```r
valid_cgndb <- unlist(lapply(regions, function(region){
  setdiff(names(d_AQHI[[region]]), c("Date", "Hour (UTC)")) #generate a list of cgndb values
}))
```

We need to adjust that data to correspond with different CMAs.

```r
CMA <- st_read(file.path(main_dir, "Data/Spatial/CMA_shape.shp")) #import CMA
```

```
Reading layer `CMA_shape' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\Spatial\CMA_shape
  using driver `ESRI Shapefile'
Simple feature collection with 156 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 3856867 ymin: 687427.7 xmax: 8996148 ymax: 3518467
Projected CRS: NAD83 / Statistics Canada Lambert
```

```r
PR <- st_read(file.path(main_dir, "Data/Spatial/PR_shape.shp")) #import PR
```

```
Reading layer `PR_shape' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\Spatial\PR_shape
  using driver `ESRI Shapefile'
Simple feature collection with 13 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 3689321 ymin: 659305 xmax: 9015751 ymax: 5242009
Projected CRS: NAD83 / Statistics Canada Lambert
```

```r
PR <- st_simplify(PR, dTolerance = 10000) #simplify PR

CMA <- CMA |> filter(PRUID != "24") #filter out QC

naps <- st_transform(naps, 3347) #CRS correction
```

```r
naps_CMA <- naps |> filter(lengths(st_intersects(naps, CMA)) > 0) #create a naps dataframe tl
naps_CMA <- st_join(naps_CMA, CMA["CMANAME"], join = st_intersects) #add CMA value

naps_df <- as.data.table(naps_CMA)[, .(cgndb, CMANAME, name, longitude = st_coordinates(geome

naps_df <- naps_df[naps_df$cgndb %in% valid_cgndb] #remove irrelevant stations

naps_df[, name := lapply(name, function(x) fromJSON(x)$en)] #adjust names to english

cgndb_key <- setNames(naps_df$name, naps_df$cgndb) #create a key of place names and cgndb val
```

```r
for(region in regions){
  loc_names <- valid_cgndb[valid_cgndb %in% names(d_AQHI[[region]])] #create a list of names
  loc_key <- as.character(cgndb_key[loc_names]) #add the key to the list, make it chr (avoidi
  setnames(d_AQHI[[region]], loc_names, loc_key) #set new names for d_AQHI!
}
```

```r
placenames <- unlist(lapply(regions, function(region){
  setdiff(names(d_AQHI[[region]]), c("Date", "Hour (UTC)")) #making a list of placenames now
}))

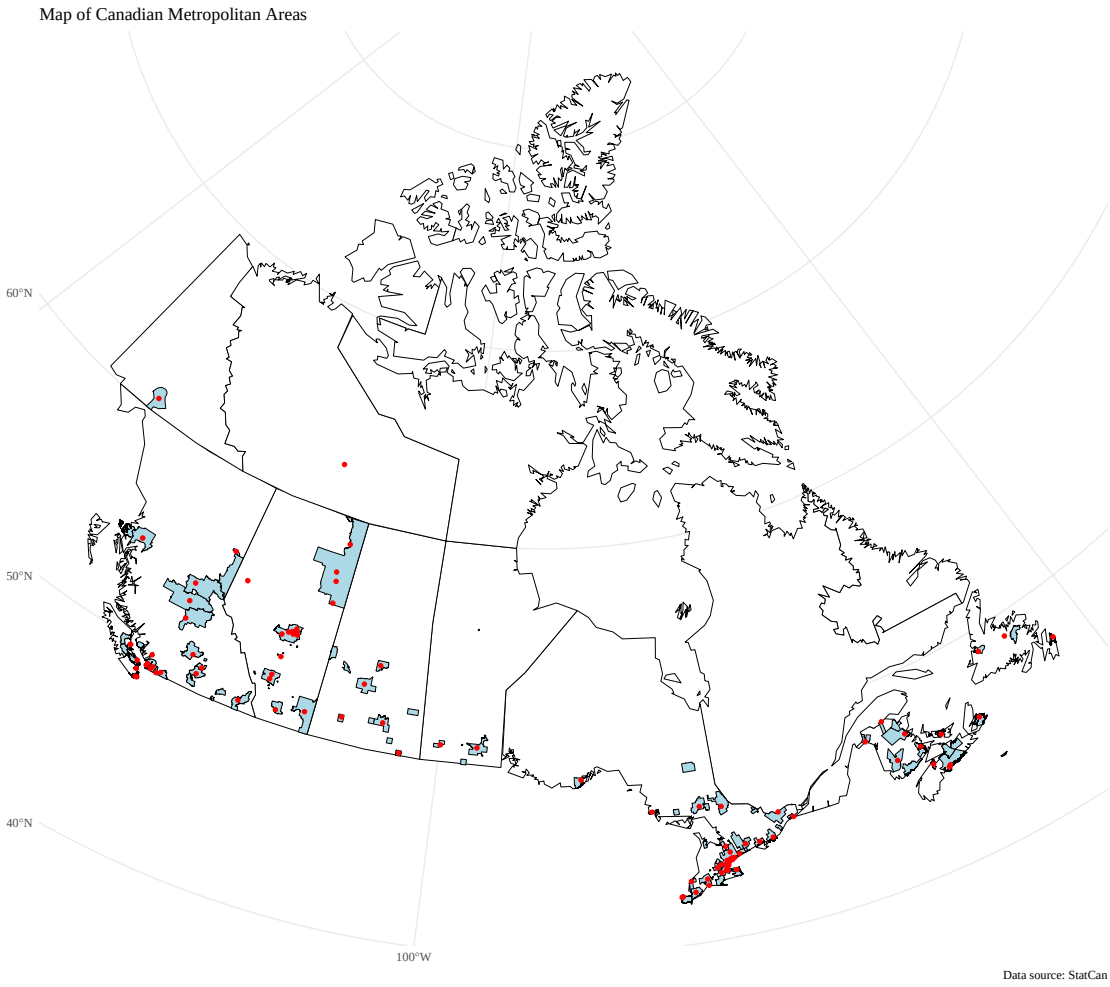
print(placenames) #check your work
```

```
  [1] "NULL"                   "Corner Brook"           "St. John's"
  [4] "Grand Falls - Windsor" "Charlottetown"          "Halifax Downtown"
  [7] "Kentville"              "Lake Major"             "Sydney"
 [10] "Halifax"                "Moncton"                "Fredericton"
 [13] "Campbellton"            "Edmundston"             "Miramichi"
 [16] "Barrie"                 "Brampton"               "Brantford"
 [19] "Burlington"             "NULL"                   "Guelph"
 [22] "Hamilton Downtown"      "Hamilton Mountain"      "London"
 [25] "Cornwall"               "North Bay"              "Oakville"
 [28] "Port Stanley"           "Toronto West"           "Windsor West"
 [31] "Sudbury"                "Thunder Bay"            "Toronto Downtown"
 [34] "Windsor Downtown"       "Hamilton West"          "Windsor"
 [37] "Newmarket"              "Peterborough"           "Milton"
 [40] "St. Catharines"         "Oshawa"                 "Toronto East"
 [43] "Toronto North"          "Sault Ste. Marie"       "Mississauga"
 [46] "Sarnia"                 "Kitchener"              "Chatham"
 [49] "Toronto"                "Kingston"               "Belleville"
 [52] "Hamilton"               "Ottawa"                 "Brandon"
 [55] "NULL"                   "Winnipeg"               "Saskatoon"
 [58] "Regina"                 "Swift Current"          "Prince Albert"
 [61] "Estevan"                "Sturgeon County"        "Strathcona County"
 [64] "Edmonton"               "Lethbridge"             "Red Deer"
 [67] "Fort Chipewyan"         "Fort McKay"             "Fort McMurray"
 [70] "Fort Saskatchewan"      "Genesee"                "St. Albert"
 [73] "Grande Prairie"         "Medicine Hat"           "Calgary"
 [76] "Wood Buffalo - south"   "Airdrie"                "Yellowknife"
 [79] "Central Fraser Valley" "Kamloops"               "Central Okanagan"
 [82] "Fort St. John"          "Comox Valley"           "Cranbrook"
 [85] "Nanaimo / Parksville"   "Metro Vancouver - SE"   "Duncan"
 [88] "Metro Vancouver - SW"   "Prince George"          "NULL"
 [91] "Quesnel"                "North Okanagan"         "Victoria / Saanich"
 [94] "Metro Vancouver - NW"   "Williams Lake"          "Terrace"
 [97] "Metro Vancouver - NE"   "WestShore"              "Comox"
[100] "Eastern Fraser Valley" "Squamish"               "Whitehorse"
```

Let's map this data.

```
ggplot() +
  geom_sf(data = PR, fill = "white", color = "black") + #PR shapes
  geom_sf(data = CMA, fill = "lightblue", color = "black") + #CMA shapes
  geom_sf(data = naps_CMA, color = "red", size = 1.2) + #naps points
  theme_minimal(base_family = "serif") +
```

```
  labs(title = "Map of Canadian Metropolitan Areas",
       caption = "Data source: StatCan") +
  coord_sf(xlim = c(3690800, 9012300), ylim = c(687400, 5242000)) #set bounds to make sure ma
```



Now let's import our census data.

```
d_Census <- fread(file.path(main_dir, "Data/Census/CMA_Data.csv")) #import census data
```

Let's take a look at the data we have available.

```
print(colnames(d_Census))
```

```
[1] "Name"
```

```
 [2] "Census Metro Area ID"
 [3] "% Household Population by Citizenship | Household Population For Citizenship | Total Ca
 [4] "% Household Population by Visible Minority Status | Household Population For Visible Mi
 [5] "# Basics | Total Population, 2024"
 [6] "Households by Income (Current Year) | Median Household Income (Current Year $), 2024"
 [7] "Households by Income (Current Year) | Average Household Income (Current Year $), 2024"
 [8] "Households by Income (Current Year) | Aggregate Household Income (Current Year $), 2024
 [9] "% Household Population 15 Years or Over by Marital Status | Household Population 15 Yea
[10] "% Household Population 15 Years or Over by Labour Force Activity | Household Population
[11] "% Male Population by Age | Males, 2024"
[12] "Total Population by Age | Average Age Of Total Population, 2024"
[13] "% Household Population 15 Years or Over by Educational Attainment | Household Populatio
[14] "% Household Population by Mother Tongue | Household Population For Mother Tongue | Tota
[15] "% Household Population by Mother Tongue | Household Population For Mother Tongue | Tota
[16] "% Household Population by Mother Tongue | Household Population For Mother Tongue | Tota
[17] "Average Total expenditure | Household Disposable Income, 2024"
[18] "% Household Population by Income (Current Year) | Household Population 15 Years Or Over
[19] "% Households by Income (Current Year) | Total Households | Household Income $100,000 Or
[20] "% Households by Income (Current Year) | Total Households | Household Income $100,000 Or
[21] "% Households by Income (Current Year) | Total Households | Household Income $100,000 Or
[22] "# Households by Income (Current Year) | Total Households, 2024"
```